Microsoft Fabric and GitHub – The story so far for 2025



Microsoft Fabric



GitHub

Kevin Chant

Agenda

- Bio
- Intro to Microsoft Fabric
- Intro to GitHub
- Configuring GitHub for Git integration
- Using GitHub with suggested CI/CD workflow options
- CI/CD for Data Warehouses and SQL database in Fabric

Kevin Chant



- Lead Technology Advocate in the Netherlands
- Worked in IT since the days of Windows 95
- Experience in various sectors
- Various certifications, Data Platform MVP



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Microsoft Fabric

Comp analytics platform

Lake centric and open

Empower every Business user

AI Powered



Fabric walkthrough



GitHub

- Manages Application Lifecycle Management
- Popular with open-source community
- Various plans
- Large range of features

GitHub Plans (formerly products)

- GitHub Free for personal accounts
- GitHub Pro
- GitHub Free for organizations
- GitHub Team
- GitHub Enterprise
  - Cloud
  - Server

As well as repositories

- GitHub Actions
- GitHub issues
- Wikis
- GitHub Copilot

GitHub Actions terminology

- Workflow – Configurable process to run jobs (pipelines)
- Action – Reusable jobs or code.
- Variables – predefined values (configured either in or out of workflow)
- Parameters – Values entered when workflow started.
- Secrets – Encrypted values stored in GitHub.

General security considerations

- Use secrets and Azure Key Vault with GitHub Actions.
- Enable code and secret scanning.
- GitHub has article on what to do if sensitive values appear in repository.
- Recommend GitHub Enterprise for production use.
- Consider GitHub certification training.

GitHub Demo



About Microsoft Fabric Git Integration

- Allows **supported** items in a workspace to have metadata synchronized with a Git repository.
  - To be more precise a workspace synchronizes with a branch.
- Supports cloud-based versions of Azure DevOps and GitHub.
- Requires Fabric or Power BI Premium capacity.
- Items supported at various levels.

Example of supported items

- Data pipelines (preview)
- Dataflows gen2 (preview)
- Eventhouse and KQL database (preview)
- EventStream (preview)
- Lakehouse (preview)
- Mirrored database (preview)
- Notebooks
- Paginated reports (preview)
- Reflex (preview)
- Warehouses (preview)
- Reports (except reports connected to semantic models hosted in Azure Analysis Services, SQL Server Analysis Services, or reports exported by Power BI Desktop that depend on semantic models hosted in MyWorkspace) (preview)
- Semantic models (except push datasets, live connections to Analysis Services, model v1) (preview)
- Spark Job Definitions (preview)
- Spark environment (preview)
- SQL database (preview)

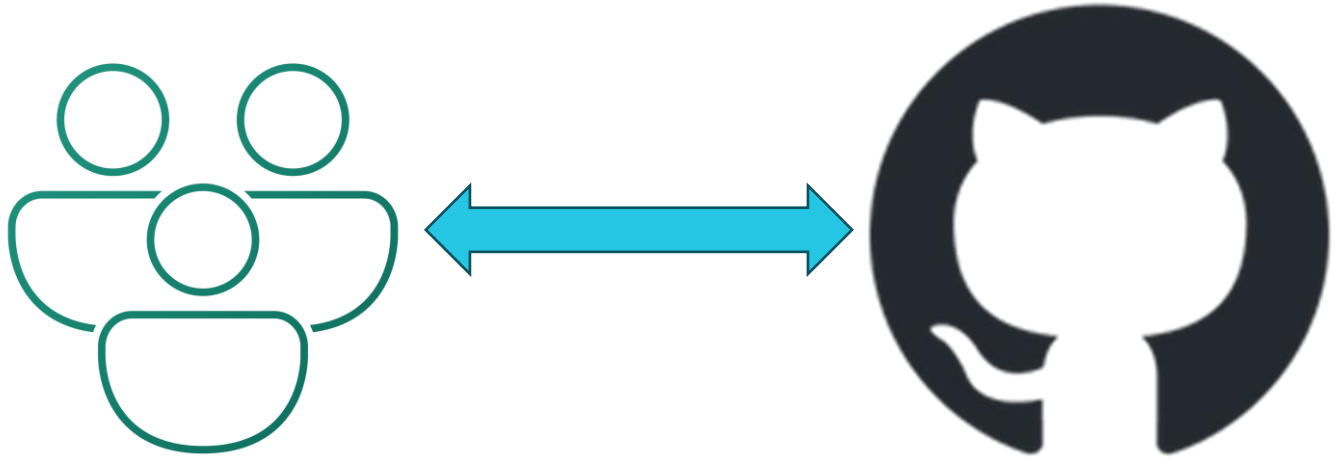
Configuring Microsoft Fabric Git Integration with GitHub

- Utilizes Personal Access Tokens.
- Unsupported items can still be deployed in the workspace.

Security considerations

- Now have ability to create GitHub organizations in EU, Australia and the US.
- Enable code and secret scanning.
- Review PAT strategy.
- Review who can manage connections and gateways in Fabric.

Git integration Demo



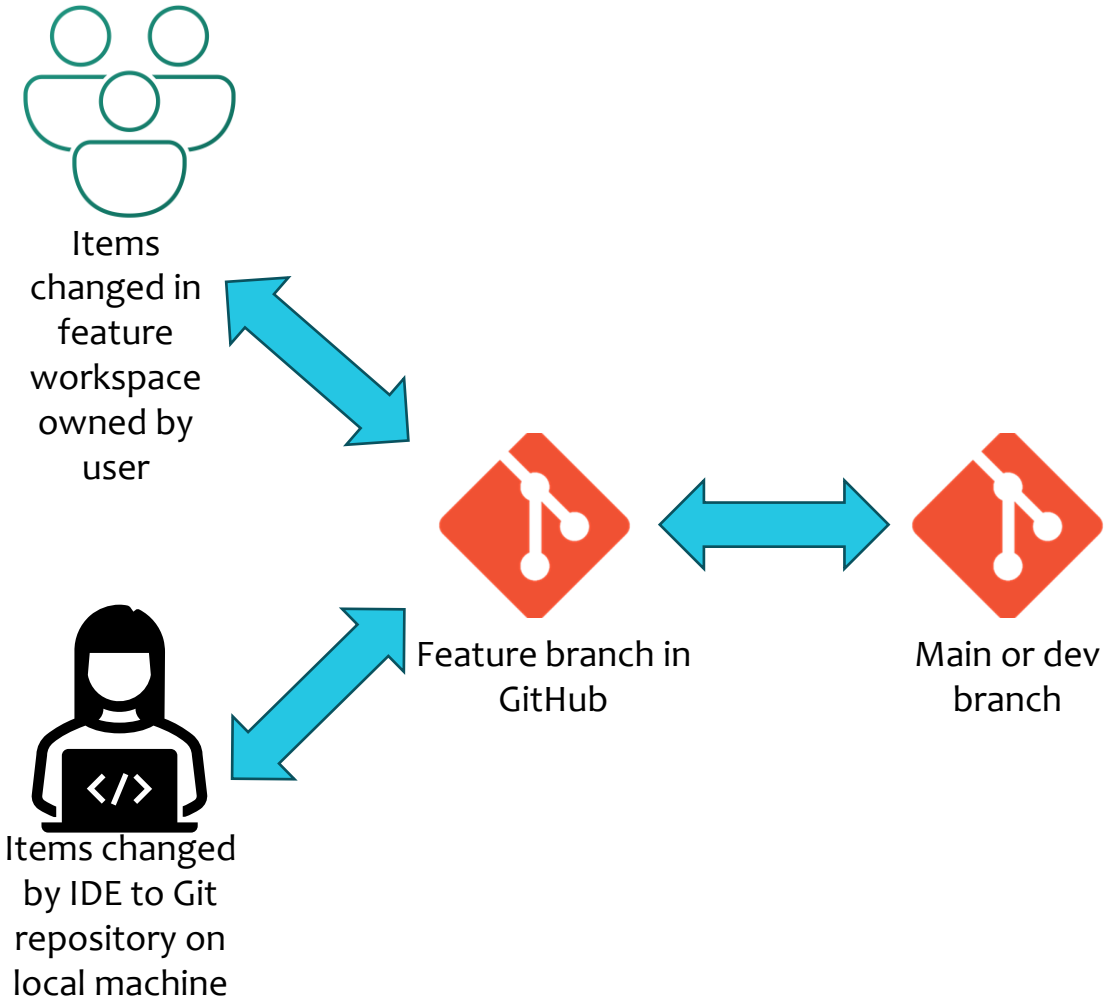
Microsoft Fabric workspace

GitHub

Using GitHub with suggested CI/CD workflow options

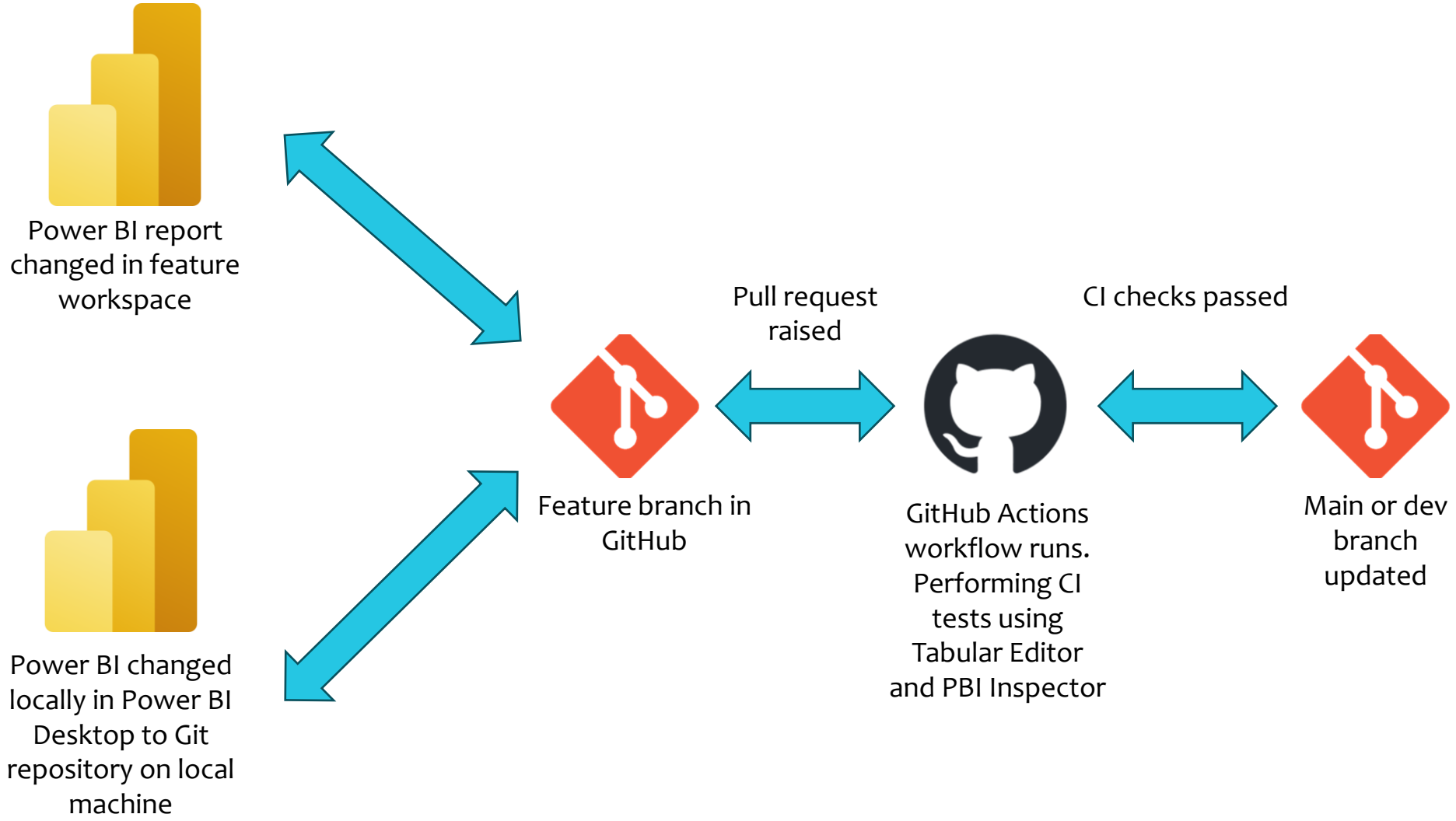
- Microsoft released article in September.
- I provide guidance how GitHub fits into suggested CI/CD workflows.

Recommended development process



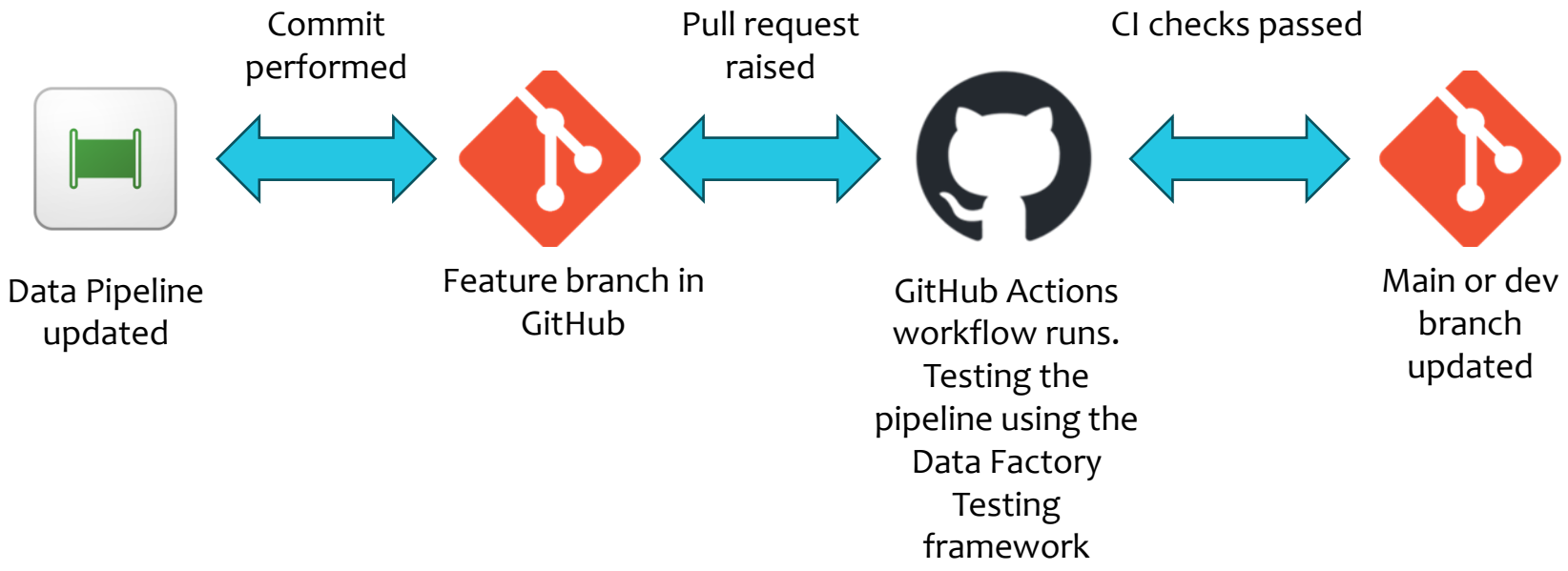
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One step further by introducing unit tests



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Testing Data Pipelines

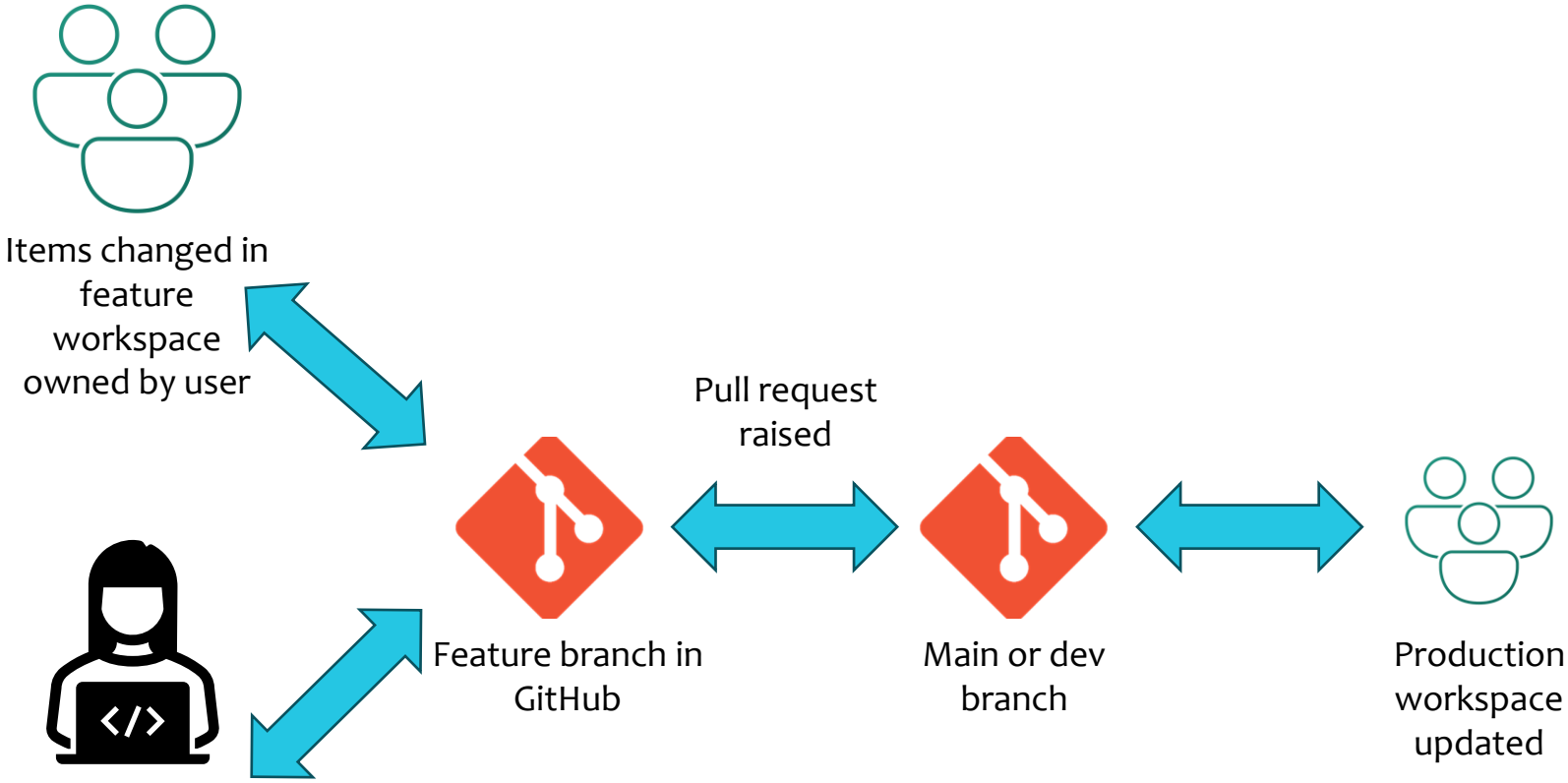


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Release Option 1 – Git-based deployments

- Deploy to multiple workspaces connected to the same Git repository.
- Achieved by having workspaces connected to different branches.

Release Option 1 - Diagram



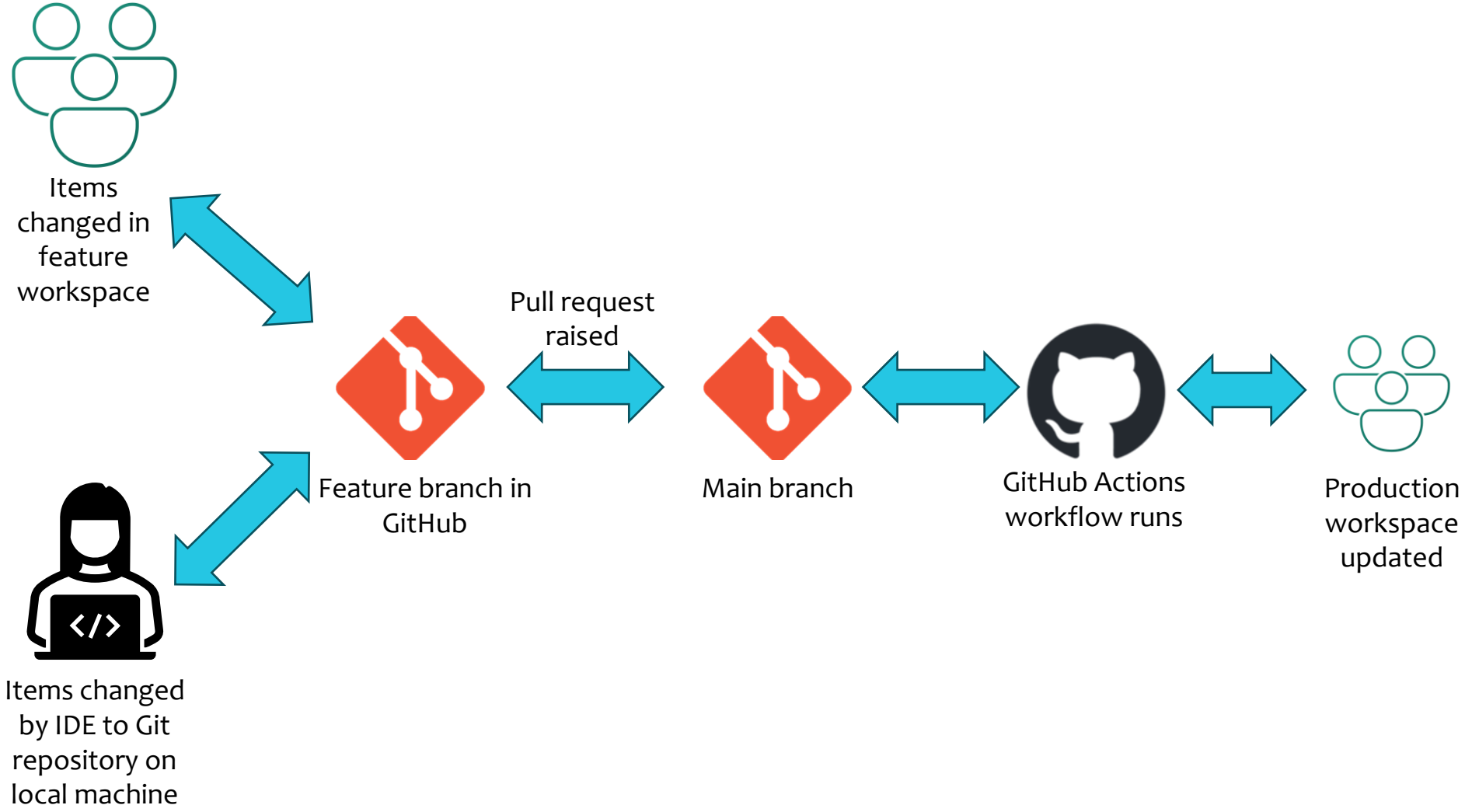
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Release Option 2 – Git-based deployments using build pipeline

- Deploy to different workspaces using GitHub Actions workflows.

- Recommendation is that each workflow contains a build and release process.
  - Build for unit tests.
  - Release to perform update.

Release Option 2 - Diagram



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Option 2 recommendations

- Avoid sensitive values in workflows. Store them in either:
  - GitHub Action Secrets.
  - Azure Key Vault (preferred).

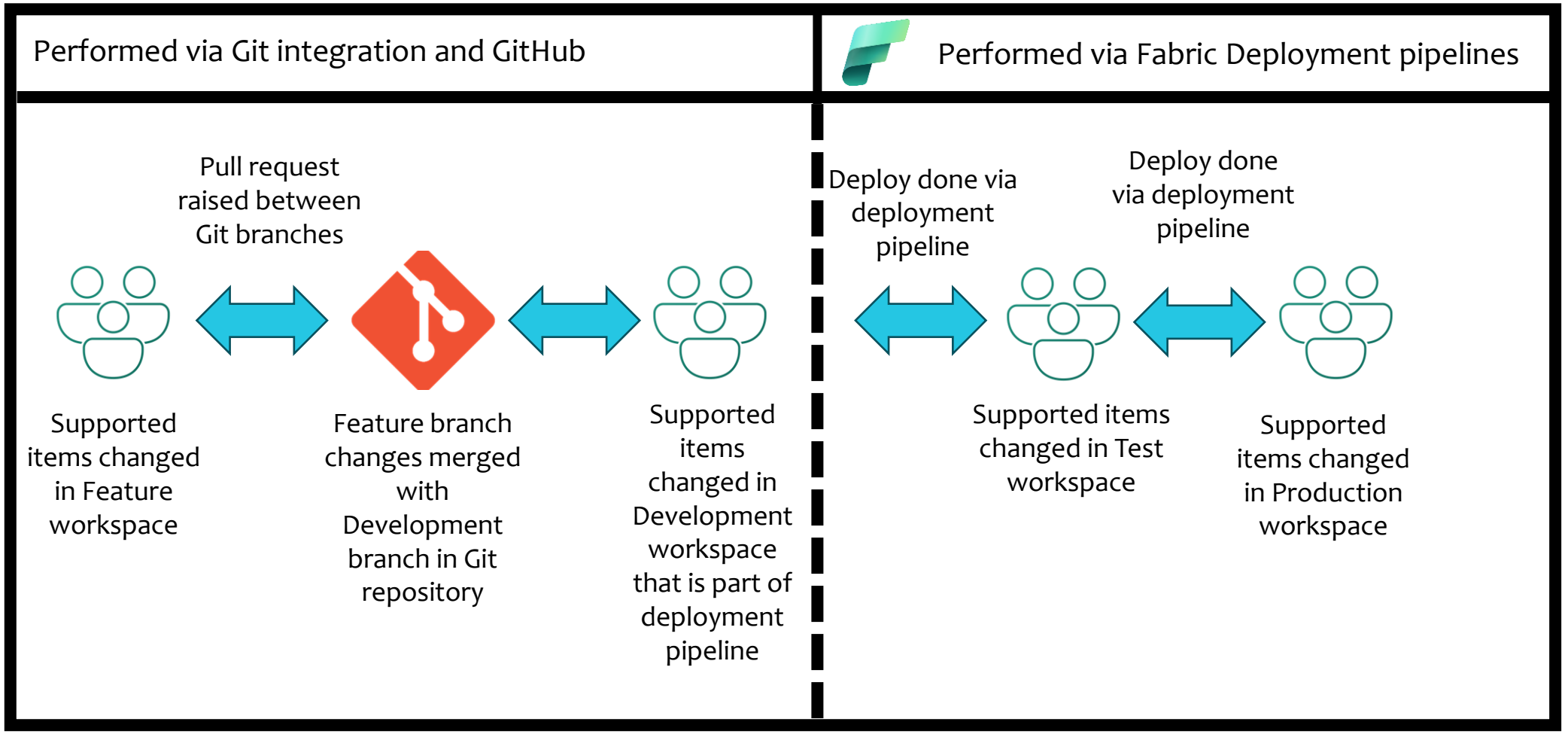
- Implement approvals process for production workloads.

- Suggest looking at tooling as layer of abstraction:
  - PowerShell modules.
  - Fabric-cicd Python library.
  - Fabric Command Line Interface.

Release Option 3 – Deploy using Microsoft Fabric deployment pipelines

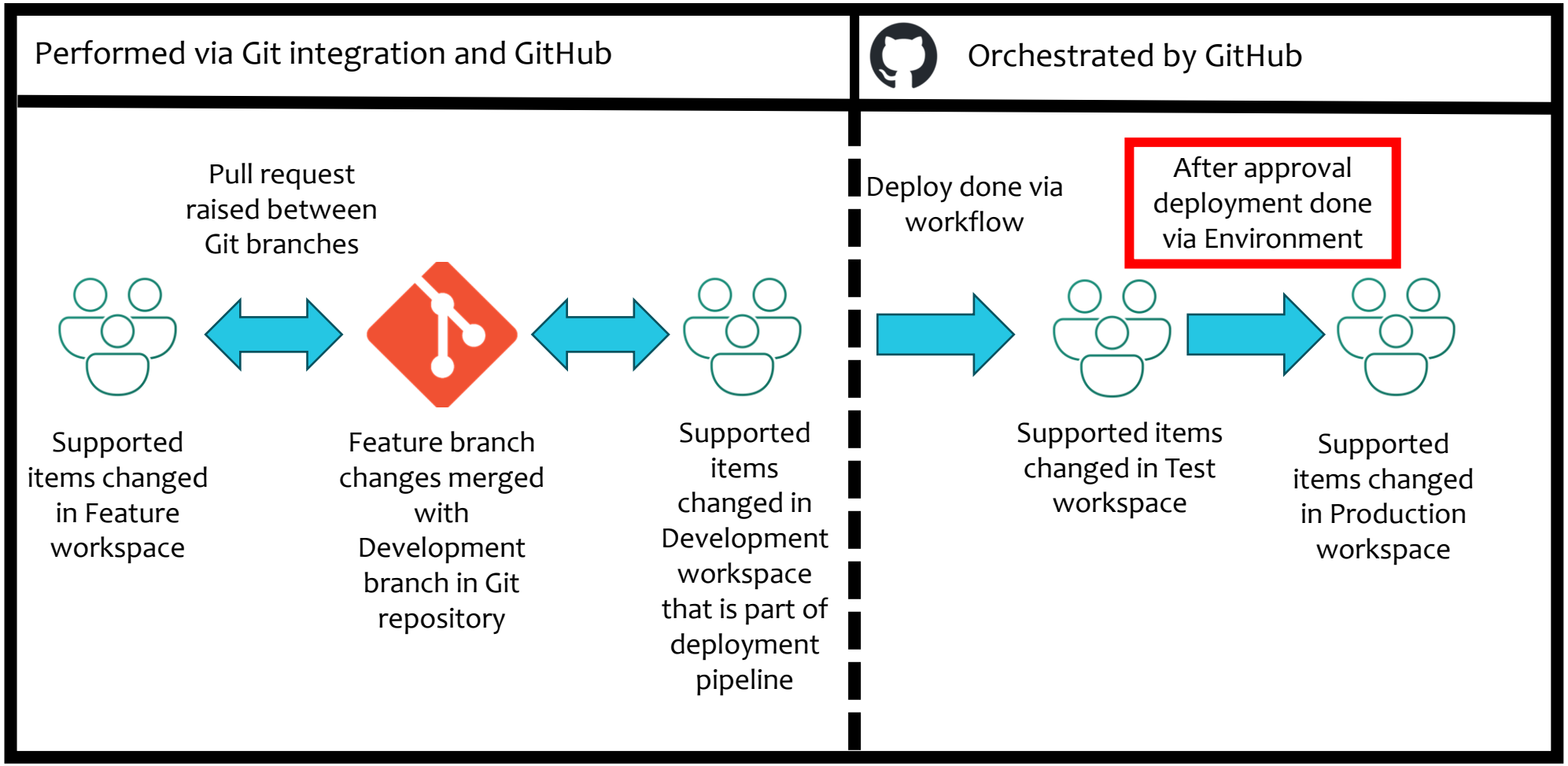
- Perform your pull request from feature branch to branch connected to a workspace that represents Dev stage of a deployment pipeline.
- From there orchestrate using Microsoft Fabric deployment pipelines.
- Alternatively, orchestrate to different Microsoft Fabric deployment pipeline stages using GitHub.

Release Option 3 - Diagram



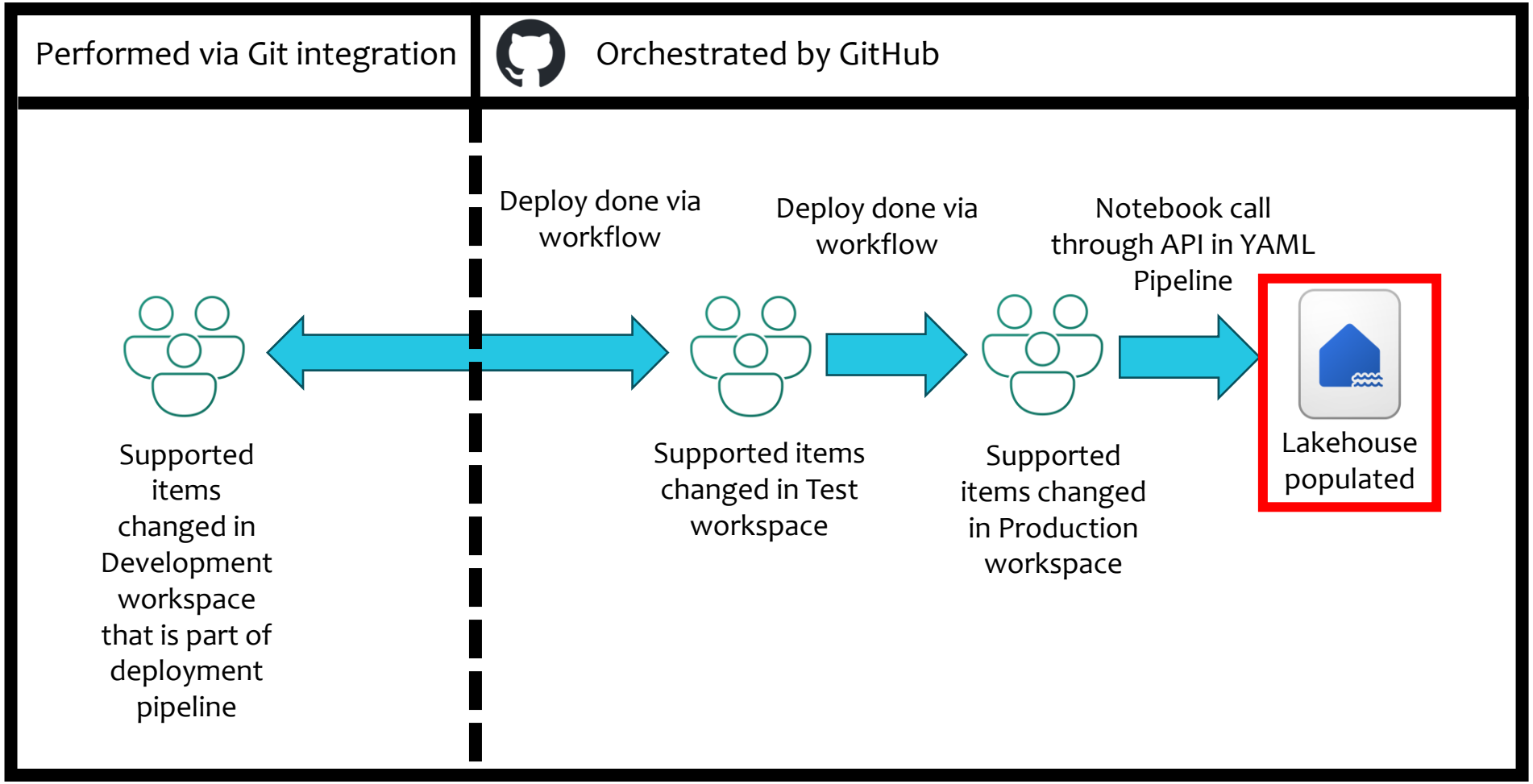
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Orchestrated by Azure Pipelines



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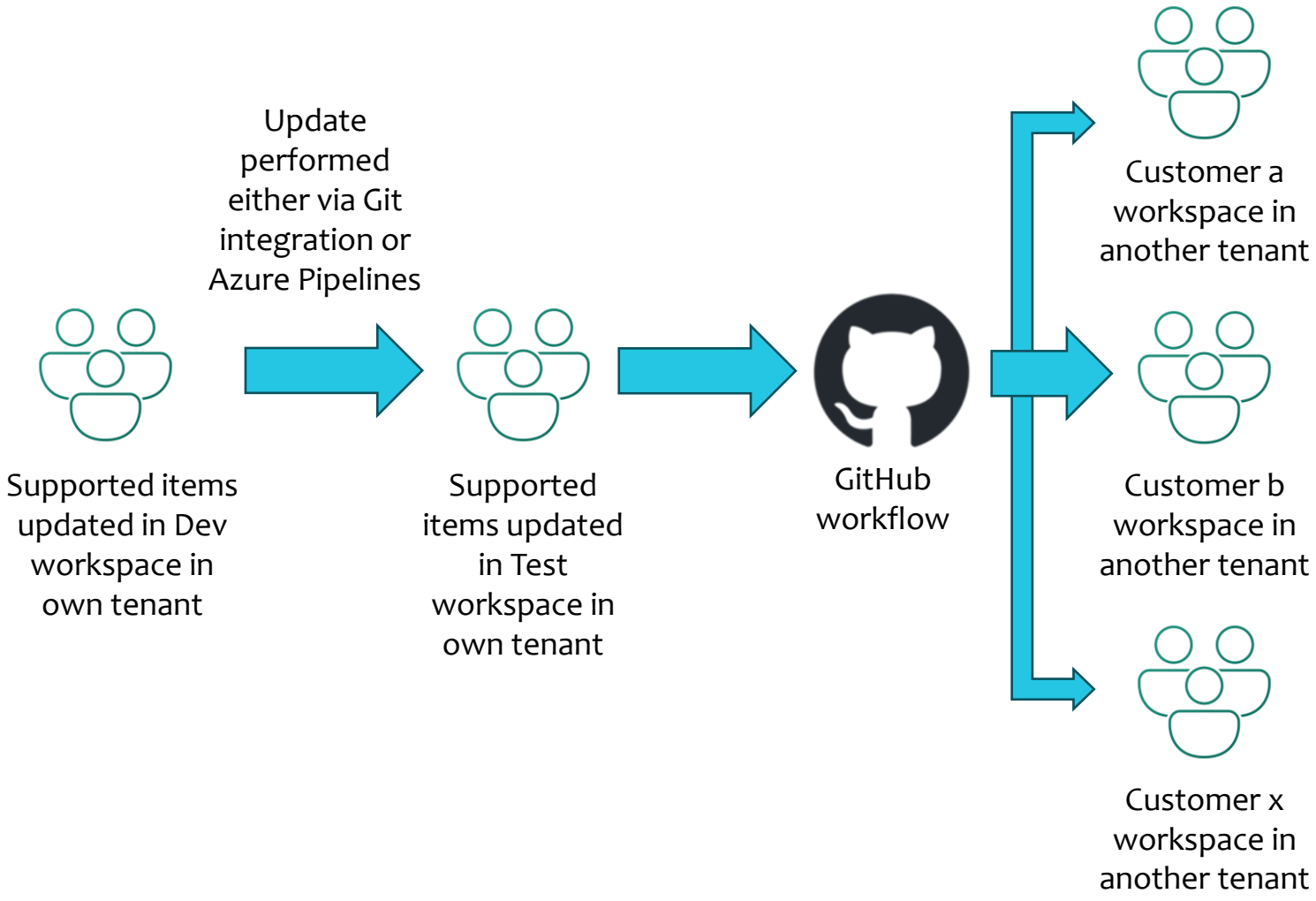
Another advantage of GitHub



Release Option 4 – For multiple customers/solutions/tenants

- Similar to option 2.
- Dev and test stages are managed in same Fabric tenant.
- Deployment to Prod stages to workspaces in other tenants using GitHub workflows.

Release Option 4 - Diagram



Demos

- Deploying to multiple workspaces via Git integration.
- Deploying via deployment pipelines.
- Deploying with GitHub workflow.
- Deploying to another tenant.

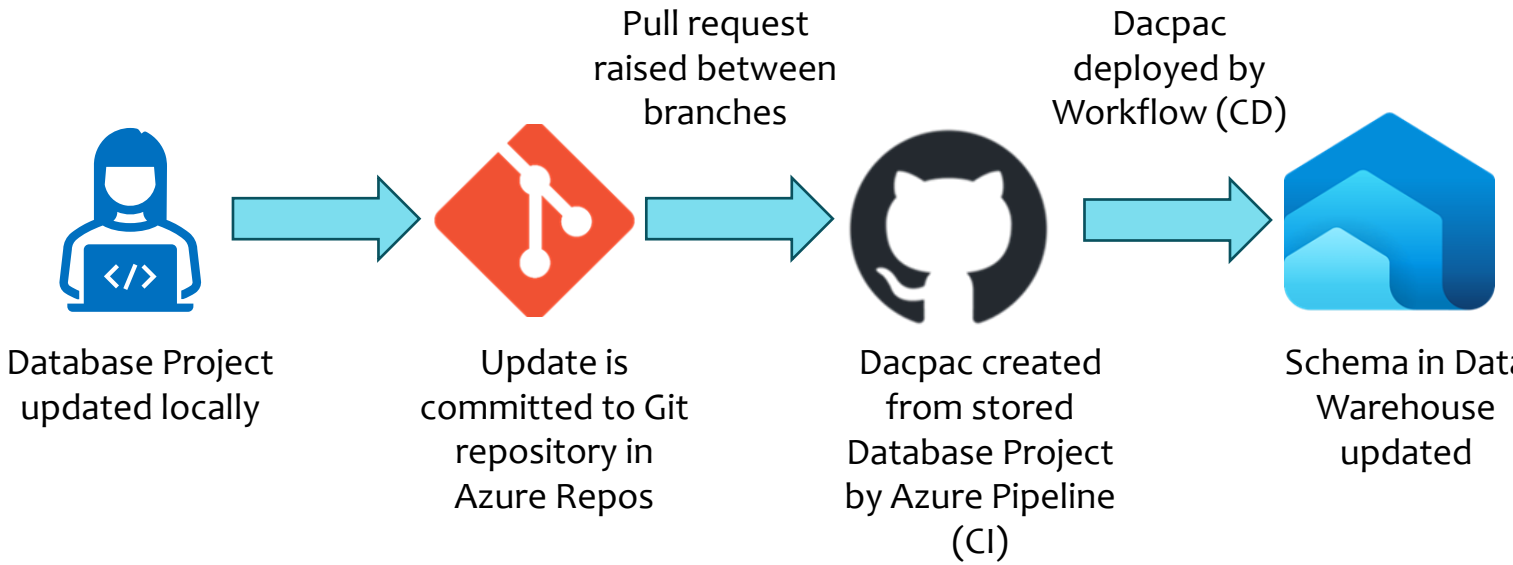
CI/CD methods for Data Warehouses and SQL databases in Fabric

- Can connect to Data Warehouse and SQL database in Fabric via connection string
- Allows state-based deployments via database projects.
- Supports Database Projects created in number of places.
- CI/CD can be performed with GitHub workflows.

Why alternative CI/CD?

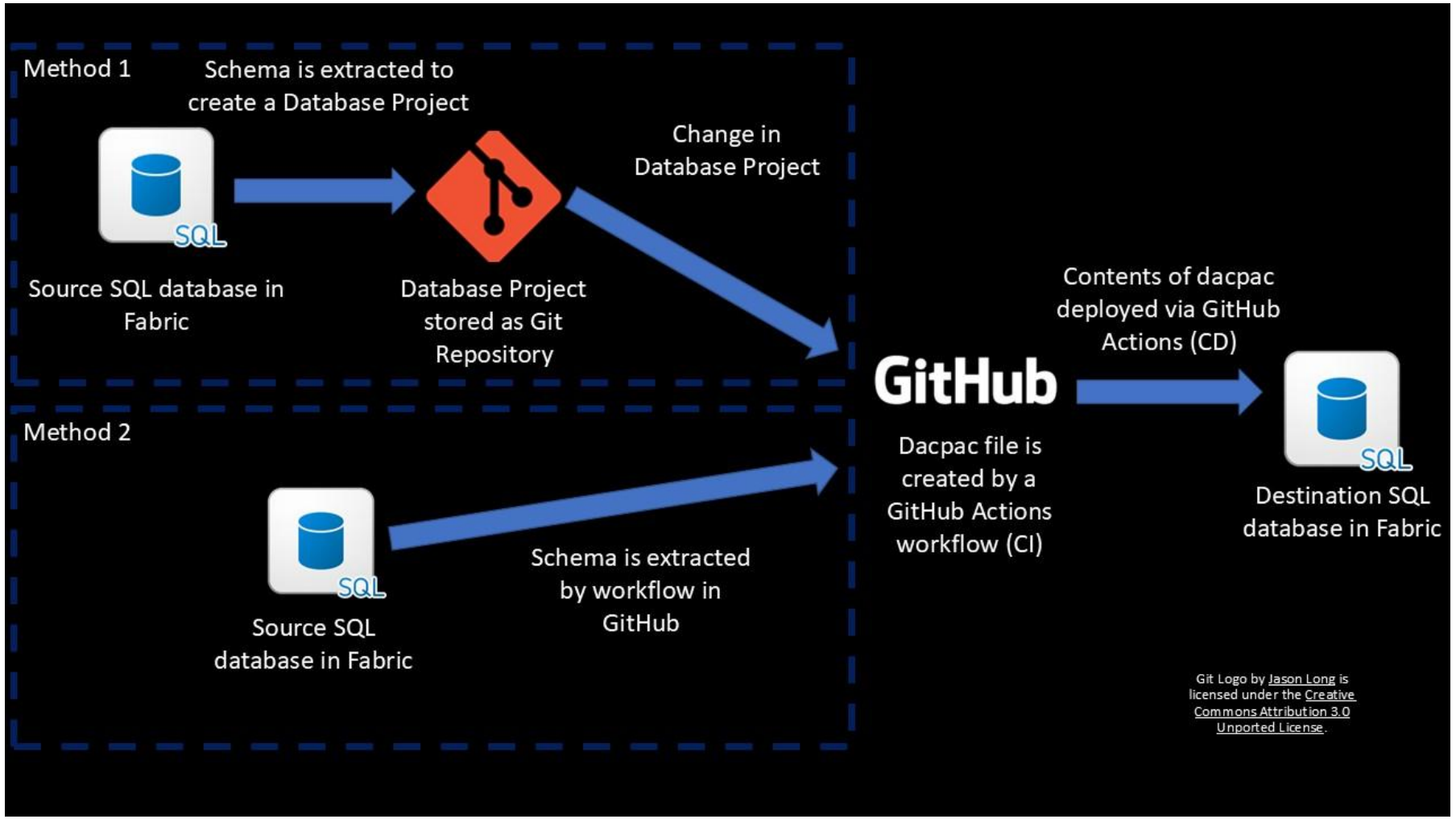
1. Git integration/Fabric deployment pipelines not implemented.
2. Multiple items in same workspace.
3. Deployments from another tenant.
4. Keep schema updates isolated.

CI/CD for Data Warehouses



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CI/CD for SQL databases in Fabric



Demo

- CI/CD for SQL database in Fabric.

To summarize

- GitHub becoming more viable for Fabric CI/CD requirements.
- For best coverage of items implement Git integration.
- Various release options available for multiple needs.
- Traditional SQL CI/CD still an option.

Questions



Thank you



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